

Chapter 18

Two-port and multiport storage elements

PREFACE

The previous chapter demonstrated the necessity of the multiport storage element for the description of energy storage in an (ideal) gas. For a constant amount of gas, for example in a cylinder–piston combination, only two ports were required. This resulted in the possibility to extend the concept of two-port transduction to transduction by means of a cycle process. Various cycle processes will be elaborated in the next chapter. This chapter first treats the storage of energy in a multiport storage element from a theoretic point of view. This results in a number of constraints for the constitutive relations of a multiport storage element. A number of transformations of the energy function are introduced, being different forms of co-energy, free energy, enthalpy and Gibbs' free energy. These transformations and constraints provide the tools to clearly describe and understand cycle processes and energy storage in multiport storage elements under various environmental conditions.

OBJECTIVES

After studying these notes the reader is expected to be able to

- recognize two- and multiport storage in real systems as a relevant phenomenon
- express the stored energy in a multiport storage element in the correct variables
- perform a Legendre transform of an energy function
- transform models containing two-port storage elements in computable equations
- represent a cycle process graphically in the port characteristics
- define Gibbs' free energy
- define enthalpy
- check Maxwell reciprocity of a multiport storage element.

STUDY MATERIAL

1 **Introduction**

Until the previous chapter storage was described by one-ports in a framework of variables based on mechanics. This chapter will approach the concept of storage from a thermodynamic point of view. Although this is a common belief, this approach does *not* violate the mechanical approach. Only the nature of the constitutive relations will appear to be different. As announced already, *transduction in storage elements* will be involved. The transducers encountered so far were power continuous transformers and gyrators. After a closer look it appears that during transduction storage processes play a role that cannot be neglected in some cases.

ASSIGNMENT 18.1

Which condition is required for a lever in order to prevent storage during the transduction process? Is this condition realistic?

ASSIGNMENT 18.2

Which non-thermal transducer have you encountered already in which storage actually cannot be neglected?

A number of extensions of storage elements and of storage components already encountered in part 1 is considered in order to get an impression of the way in which two- and multiport storage elements can be important even when the thermal domain plays no role. Consider an electric capacitor, a water container and a coil. In all cases a situation where a one-port storage element is a competent model is considered first. By allowing the construction of the component to vary one of configuration parameters becomes a state variable. The force that is the conjugated variable of this displacement can be derived from the stored energy by differentiation. In the previous chapter an example was already encountered:

$$F = \frac{dE}{dx} \quad (18.1)$$

or, if E does not only depend on x , by partial differentiation:

$$F = \frac{\partial E}{\partial x} \quad (18.2)$$

After discussing the examples of the capacitor, the coil and the water container, the description is generalized (made domain independent) by abstraction. A close relation will be found with the formalizing work in the previous century by *Gibbs* and by *Maxwell* in thermodynamics, but also with the work by the Irish mathematician and astronomer *Sir William Rowan Hamilton* (1805–1865) some decades earlier. The chosen causality of the storing ports will appear to be closely connected with so-called *Legendre transforms*.

Hamilton united optics and mechanics and unconsciously founded in 1835 quantum mechanics with his work ‘On a general method in dynamics’. It contains a careful formulation of the concept of *energy* on the basis of the actually stored quantities. Only after quantum mechanics emerged some hundred years later, the deep meaning of his work was fully appreciated. Legendre transforms are named after the French mathematician *Adrien-Marie Legendre* (1752–1833). Due to the envy of his colleague *Pierre-Simon Laplace* (1749–1827), who’s name was already encountered via the Laplace transform, he never received an appropriate appreciation nor position in spite of the fact that his German contemporary and one of the most important (applied) mathematicians of that time, *Carl Friedrich Gauss* (1777–1855), considered him as the ‘gem of arithmetic’.

2 Examples of two-port storage elements

2.1 FLAT-PLATE CAPACITOR WITH A MOVING PLATE

The first example is the flat-plate capacitor discussed in chapter 3 (figure 3.1). Assuming a linear constitutive relation $u = q/C$, where $C = A\epsilon/d$ the configuration is changed such, that one plate can move in the normal direction with respect to the other (figure 18.1).

A capacitor with a moving plate can be used as a microphone.

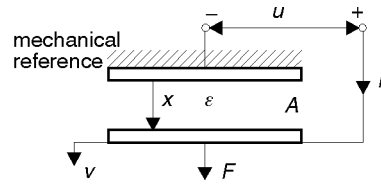


FIGURE 18.1
Capacitor with moving plate

This change the parameter of the distance d into a variable to be called x . Apart from the charge q on the plates, this variable x is also required to fix the state of this system. Hence, there is a second state variable. Accordingly, the stored energy becomes a function of both charge q and distance x between the plates:

$$E(q, x) = \frac{q^2}{2C(x)} = \frac{q^2 x}{2A\epsilon} \quad (18.3)$$

In the general case of a one-port q -type storage element it holds that:

$$dE = edq \quad (18.4)$$

and for the effort e :

$$e = \frac{dE}{dq} \quad (18.5)$$

If the energy content of the capacitor depends not only on the charge q but also on the distance between the plates x a two-port storage element is involved. In case of a two-port the effort of each of the ports can be determined by means of partial derivatives:

$$u = \left. \frac{dE}{dq} \right|_{dx=0} = \frac{\partial E}{\partial q} = \frac{qx}{A\epsilon} = \frac{q}{C(x)} \quad (18.6)$$

and

$$F = \left. \frac{dE}{dx} \right|_{dq=0} = \frac{\partial E}{\partial x} = \frac{q^2}{2A\epsilon} \quad (18.7)$$

F represents the force that needs to be applied to the plates in order to maintain a displacement x . Note that this force does not depend on x , the distance between the plates. Obviously, this distance cannot become so large that fringing effects violate the assumption of a linear constitutive relation.

Also note that it is not necessary to derive this force via an analysis of the forces between the charges. However, the single linear constitutive relation of the one-port is turned into two *non*-linear relations for the two-port. This means that the concepts of ‘capacitance of the electric port’ and ‘spring constant of the mechanical port’ are not valid anymore.

For a ‘common’ capacitor that can be modeled by a one-port storage element, it holds that:

This will be treated in a more formal way in section 6.

$$u = \frac{q}{C} \rightarrow \frac{du}{dq} = \frac{1}{C}$$

and in the same way for a ‘common’ spring:

$$F = \frac{x}{C} \rightarrow \frac{dF}{dx} = \frac{1}{C}$$

Such relations can be found also in case of a multiport storage element. However, *partial* derivatives are required, because u and F depend on both q and x in principle. For example:

$$\frac{\partial u}{\partial q} = \frac{x}{A\epsilon} \quad \text{and} \quad \frac{\partial u}{\partial x} = \frac{q}{A\epsilon}$$

All partial derivatives of $u(q,x)$ and $F(q,x) = F(q)$ with respect to q and x can be determined and written in matrix form:

$$\frac{\partial(u, F)}{\partial(q, x)} = \begin{pmatrix} \frac{\partial u}{\partial q} & \frac{\partial F}{\partial q} \\ \frac{\partial u}{\partial x} & \frac{\partial F}{\partial x} \end{pmatrix} = \begin{pmatrix} \frac{x}{A\epsilon} & \frac{q}{A\epsilon} \\ \frac{q}{A\epsilon} & 0 \end{pmatrix} \quad (18.8)$$

such that near the point (q_0, x_0) :

$$\begin{pmatrix} du \\ dF \end{pmatrix} = \begin{pmatrix} \frac{x_0}{A\epsilon} & \frac{q_0}{A\epsilon} \\ \frac{q_0}{A\epsilon} & 0 \end{pmatrix} \begin{pmatrix} dq \\ dx \end{pmatrix} \quad (18.9)$$

Symmetric matrix

Note that this matrix has equal non-diagonal terms. Such a matrix is called a *symmetric matrix*. For a symmetric matrix it holds by definition that $\mathbf{A} = \mathbf{A}^T$. This will be encountered too in the other examples and reconsidered in section 4. The capacitor with a moving plate is a two-port storage element with an electric and a mechanical port. This opens the possibility of transduction from mechanical energy into electric energy vice versa.

ASSIGNMENT 18.3

Which condition needs to be satisfied to enable transduction?

ASSIGNMENT 18.4

List practical applications of the flat-plate capacitor as an electromechanical transducer.

Characteristics

Finally the *characteristics* of both ports in preferred causality are drawn, i.e. $u = u(q)|_{dx=0}$ and $F = F(x)|_{dq=0}$. A *characteristic* is the graphical representation of the *constitutive relation of a port* when the state or equilibrium

Bundle of characteristics

determining variable of the other port is kept constant. In this case $u(q)$ is sketched with x kept constant (straight line through the origin) and $F(x)$ with q kept constant (horizontal line). By sketching a characteristic for different values of the other port variable a *bundle of characteristics* is obtained. In case of the bundle of $u(q)$ the characteristics ‘pivot’ in the origin where the slope is steeper for increasing values of x_0 (figure 18.2a). In case of the bundle of F – x –characteristics F does not depend on x , which puts the pivot point at infinity and the characteristics in parallel. Their distance to the x –axis is larger for increasing values of $|q_0|$ (figure 18.2b).

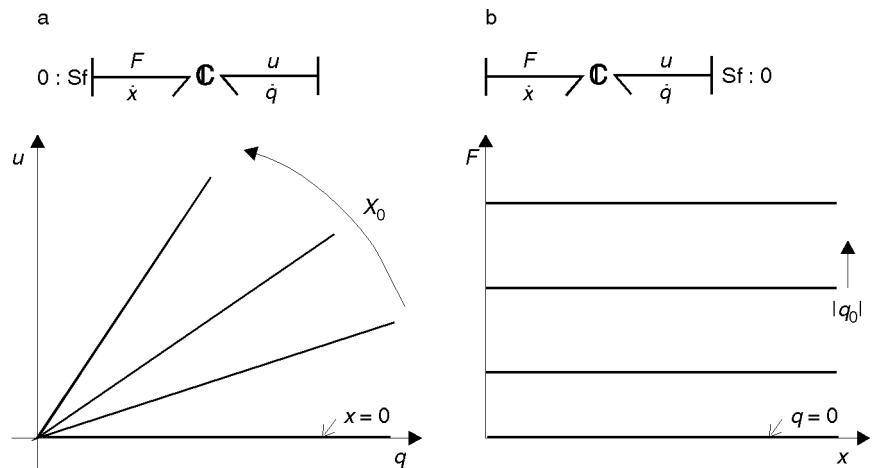


FIGURE 18.2 Bundles of characteristics of
 a $u(q, x_0) = u(q) \big|_{dx=0}$
 b $F(q_0, x) = F(x) \big|_{dq=0}$

ASSIGNMENT 18.5

For which values of x is the horizontal line $F(x)$ realistic? What happens to the characteristics for values of x where the horizontal line isn't realistic anymore?

ASSIGNMENT 18.6

Suppose that the environment does not apply a force on the plates while a certain amount of energy is stored in the capacitor. How will the plates move?

2.2 WATER CONTAINER WITH FLOAT

Again a known model is taken as a starting point, viz. the model of a water container with straight walls. It holds that:

$$p = \frac{V}{C} = \frac{\rho g}{A} V \quad (18.10)$$

where A [m²] is the cross section area. Next a watertight object ('float') is inserted that also has straight walls and cross section area a [m²]. Initially this float is fixed at level l [m] with respect to the bottom of the container (figure 18.3).

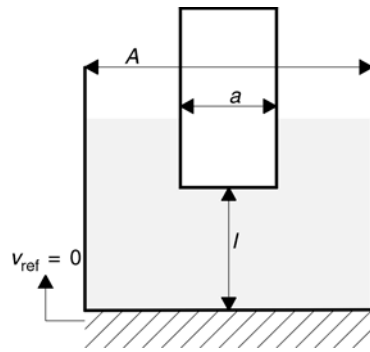


FIGURE 18.3
Water container with float

ASSIGNMENT 18.7

How does the constitutive relation of the container change by this addition?
Sketch this relation and compute the energy stored in the container, assuming an interest in the situation where the float is partially submerged.

This changes
 l into x .

After the stored energy is derived (cf. solution to assignment 18.7) the level of the float can be made variable such that a second state variable originates. It is assumed that the float is massless.

$$E(V, x) = \frac{\rho g}{2(A-a)} (V^2 - 2Vax + aAx^2) \quad \text{for } V > Ax \quad (18.11)$$

$$E(V) = \frac{\rho g V^2}{2A} \quad \text{for } V < Ax \quad (18.12)$$

The force to be exerted on the float in order to keep it at a distance x from the bottom, can be derived by partial differentiation of the expression for the energy with respect to x :

$$F = \frac{\partial E}{\partial x} = \frac{-\rho g a}{A-a} (V - Ax) \quad \text{for } V > Ax \quad (18.13a)$$

For this situation holds that: $F < 0$, i.e. the float should be pushed down. This corresponds to common experience.

$$F = 0 \text{ for } V < Ax \quad (18.13b)$$

In this case the float is not yet in contact with the water.

The energy function can be simply checked by partial differentiation with respect to V , which should recover the expression of the pressure p that served as a starting point:

$$\begin{aligned} p &= \frac{\partial E}{\partial V} = \frac{\rho g}{A-a} (V - ax) = \frac{\rho g}{A-a} (V - Ax + (A-a)x) = \\ &= \frac{\rho g}{A-a} (V - Ax) + \rho g x \quad \text{for } V > Ax \end{aligned} \quad (18.14a)$$

and

$$p = \frac{\rho g V}{A} \quad \text{for} \quad V < Ax \quad (18.14b)$$

In $(V - Ax)/(A - a)$ the distance can be recognized by which the float is submerged. Accordingly, the volume of the *displaced water* is $a(V - Ax)/(A - a)$, such that

$\rho g a \frac{V - Ax}{A - a}$ is the *weight of the displaced water*. Consequently, the force to be

exerted on the float is equal in value, but oppositely directed to the buoyancy force according to *Archimedes' buoyancy law*. In this case the linear description in matrix form of small variations around the state (V_0, x_0) is retrieved by partial differentiation of the pressure p and the force F with respect to the volume of water V and the float position x . Only the case is considered that the float is partially submerged:

$$\begin{pmatrix} dp \\ dF \end{pmatrix} = \begin{pmatrix} \frac{\rho g}{A - a} & \frac{-a \rho g}{A - a} \\ \frac{-a \rho g}{A - a} & \frac{a A \rho g}{A - a} \end{pmatrix} \begin{pmatrix} dV \\ dx \end{pmatrix} \quad (18.15)$$

Note that the matrix is symmetric, just like the constitutive matrix of the flat-plate capacitor with moving plate.

ASSIGNMENT 18.8

Sketch the bundles of characteristics of the both ports of the water container with float.

Archimedes' law

Archimedes



2.3 ELASTIC COIL

The final example of a two-port storage element considered here is a spiral spring of conducting material, e.g. copper wire, that can also be part of an electric circuit as a coil (figure 18.4).

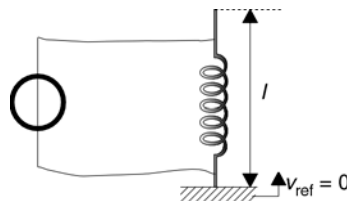


FIGURE 18.4
Spring used simultaneously as coil

Only the storage processes are considered, viz. the storage of energy via the electric —or rather *magnetic* (cf. chapter 3)— port and via the *mechanical* port. The magnetic port is taken as a starting point again and also assumes that the coil has a linear constitutive relation with self induction L

$$i = \frac{\lambda}{L} = \frac{\lambda l}{A \mu} \quad (18.16)$$

where l [m] is the length of the coil and A [m²] its cross section area.

The energy E [J] is

$$E(\lambda) = \frac{\lambda^2}{2L} = \frac{\lambda^2 l}{2A\mu} \quad (18.17)$$

If the length of the coil is varied next again, the constant l becomes the (state) variable x . Consequently, the coil is also used as spring. It is found that the spring stores energy too of the coil is without current, i.e. if $\lambda = 0$. Again a linear relation between force F and extension or displacement $(x - l)$ is assumed, according to Hooke's law

$$F = K(x - l) \quad (18.18)$$

corresponding to the elastic energy

$$E(x) = \frac{K(x - l)^2}{2} \quad (18.19)$$

such that the total energy of an elastic coil $E(\lambda, x)$ equals

$$E(\lambda, x) = \frac{\lambda^2 x}{2A\mu} + \frac{K(x - l)^2}{2} \quad (18.20)$$

Partial differentiation yields the expression for the current through the elastic coil

$$i(\lambda, x) = \frac{\partial E(\lambda, x)}{\partial \lambda} = \frac{\lambda x}{A\mu} = \frac{\lambda}{L(x)} \quad (18.21)$$

as well as an expression for the force at the elastic coil

$$F(\lambda, x) = \frac{\partial E(\lambda, x)}{\partial x} = \frac{\lambda^2}{2A\mu} + K(x - l) \quad (18.22)$$

The first term is, just like in case of the flat-plate capacitor, independent of x , but in this case there is a second term describing the action of a spring according to Hooke's law.

ASSIGNMENT 18.9

In which case would the description of the flat-plate capacitor also have contained a similar term?

Just like in case of the first two examples the linear description in matrix form of small variations around the state (λ_0, x_0) is derived by partial differentiation of the current i and the force F with respect to the flux linkage λ and the extension of the spring x :

$$\begin{pmatrix} di \\ dF \end{pmatrix} = \begin{pmatrix} \frac{x_0}{A\mu} & \frac{\lambda_0}{A\mu} \\ \frac{\lambda_0}{A\mu} & K \end{pmatrix} \begin{pmatrix} d\lambda \\ dx \end{pmatrix} \quad (18.23)$$

ASSIGNMENT 18.10

Represent the three examples in bond graph form.

2.4 DISCUSSION OF THE EXAMPLES

The symmetry of the partial derivatives in the constitutive matrices can hardly be a coincidence. The difference between the flat-plate capacitor and the two other examples is that the lower diagonal term is not zero in the latter cases. In case of the flat-plate capacitor this causes the determinant of this matrix to be always negative, viz. $-(q/A\epsilon)^2$. In case of the container with float the determinant is $a(A-a)[\rho g/(A-a)]^2$. This determinant is positive as long as $A > a$, a necessary condition for the float to fit in the container. By the way, if $A = a$ a cylinder-piston combination is obtained, where the states V and x are not independent anymore, because the density of the water is assumed constant. This means that no compression can take place like in the cylinder-piston combination filled with gas in the previous chapter. Finally, the determinant of the elastic coil equals

$$\frac{Kx}{A\mu} - \left(\frac{\lambda}{A\mu} \right)^2$$

The sign of this determinant depends on the values of K , x and λ . Anticipating a later analysis it is predicted here already that if the spring constant is too small, i.e. the spring too soft, the windings will collapse for a certain extension and flux linkage. This causes an ‘implosion’ of the spring similar to the flat-plate capacitor. In the latter the determinant is negative. Accordingly, it will be shown that the this determinant being positive is a necessary and sufficient condition for the *intrinsic stability* of a two-port storage element. Intrinsic stability means that the system has a stable equilibrium state by itself and will not, for example, implode or explode.

Intrinsic stability

ASSIGNMENT 18.11

Determine the condition of intrinsic stability of the system consisting of a point mass m [kg] horizontally rotating with angular velocity ω [rad s⁻¹] while connected to a spring with length x and rest length l [m] and spring constant K [N m⁻¹]. Hint: take the elastic energy according to (18.19) and the kinetic energy

$$E = \frac{1}{2} \frac{b^2}{J} = \frac{1}{2} \frac{b^2}{m x^2} \text{ as a starting point.}$$

Finally it is noted for these examples that it is sometimes hard to give a two-port storage element its preferred causality in a specific model. In case of the capacitor with the moving plate usually the description of the force as a function of the *voltage* and the plate position is preferred over a function of the *charge* and the plate position, because it is easier to impose a specific voltage to a capacitor in practice than a specific charge. The constraint that the charge cannot

Bias voltage

become zero in order to maintain two-port behavior is equivalent with the constraint that the voltage should not become zero. This means that a so-called *bias-voltage* always has to be present in order to apply this principle e.g. in a microphone or in a loudspeaker. Likewise, it is easier to impose a current to a coil than a flux linkage. In case of the elastic coil the description of the force by a function of the current through the coil and of its length is preferred accordingly over a function of the flux linked to the coil and its length. In this case it holds too that the coil cannot be without flux linkage and consequently not without current in order to maintain two-port behavior.

For example, if $F(u, x)$ is determined for the capacitor by substitution of $q(x, u)$ (corresponding to (18.6)) into $F(q, x) = \partial E(q, x) / \partial x$ (corresponding to (18.7)), then

$$F(u, x) = \frac{A\epsilon u^2}{2x^2} \quad (18.24)$$

However, if $F(u, x)$ is determined from the partial derivatives of $E(u, x)$, obtained by substitution of $q(x, u)$ into $E(x, u)$ according to (18.3) it is first found that

$$E(u, x) = \frac{C(x)u^2}{2} = \frac{A\epsilon u^2}{2x} \quad (18.25)$$

If the partial derivative of $E(u, x)$ with respect to x is taken next, it is found that

$$\frac{\partial E(u, x)}{\partial x} = -\frac{A\epsilon u^2}{2x^2} = -F(u, x) \quad (18.26)$$

Note that $F(q, x)$ was derived from the partial derivative of the energy $E(q, x)$ with respect to x . If both F and E are written as a function of (u, x) , then F is *not* equal to the partial derivative of the energy $E(u, x)$ with respect to x , but equals $-F$. Obviously, $E(q, x)$ and $E(u, x)$ are not fully identical in this respect.

ASSIGNMENT 18.12

Write the force acting on the elastic coil and the energy of the elastic coil as a function of the current i and the length x and show that the force is still the partial derivative of the energy with respect to the position.

ASSIGNMENT 18.13

Express the force exerted at the float in the water container and the stored energy in terms of the pressure p at the bottom of the container and of the float position x . Does $F(p, x) = -\partial E(p, x) / \partial x$ also hold in this case?

From the above as well as the solutions of the assignments 18.12 and 18.13 it can be concluded that a careful study of energy functions is required. It has become obvious that storage element contents (q - and p -type variables) cannot be simply replaced by equilibrium determining variables (e -type and f -type variables respectively). To that end the next section will first elaborate the concepts of energy and co-energy.

3 Energy in thermodynamics: Legendre transforms and the Gibbs relation

The remaining part of this chapter will demonstrate in what way the domain independent concepts encountered so far and represented by bond graph symbols are related to concepts from thermodynamics. The previous section already showed that the constitutive relations of a multiport C-type storage element relate the effort- and q -type variables. It was also shown that the energy is a function of these q -type variables. In case the storage element has an integral causality, this is the logic choice. However, the causality of a storing port is sometimes such that the effort- rather than the q -type variable is considered the independent variable. It seems logical to express the energy in that case as a function of the effort. However, this also changes the nature of the energy function. Chapter 2, section 4.2, already showed that if e and q are exchanged, a complementary energy or *co-energy* is used. Consequently, it is useful to elaborate the relation between energy and co-energy.

3.1 ENERGY, CO-ENERGY AND LEGENDRE TRANSFORMS

Chapter 2 and 3 already showed that apart from the energy E also the co-energy E^* can be computed. As mainly linear constitutive relations were considered, this distinction did not seem to be important as it then holds that $E = E^*$. Linearity of the constitutive relations of multiport storage elements is rather exception than rule. Moreover, the situation can occur that only at one of the ports of a multiport storage element q and e need to be interchanged according to the required causality of the port. Therefore, the differences between energy and co-energy are discussed in detail. First linear constitutive relations are assumed, but the discussion holds too for a non-linear constitutive relation.

Consider a q -type storage element with constitutive relation:

$$e = \frac{q}{C} \quad (18.27)$$

The energy E stored in this storage element equals

$$E(q) = \int e dq \quad (18.28)$$

This relation can be graphically represented by the area below the curve $e = q/C$ (figure 18.5).

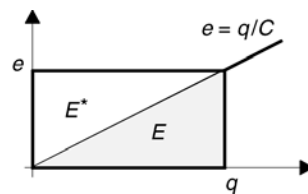


FIGURE 18.5
Energy and co-energy

The co-energy is equals

$$E^*(e) = \int q de \quad (18.29)$$

Figure 18.5 shows that the co-energy can be written too in the form

$$E^* = eq - E \quad \text{or} \quad E^* = -(E - eq) \quad (18.30)$$

where the product eq equals the area of the rectangle with sides e and q .

As

$$e = \frac{dE}{dq}$$

this can be rewritten

$$E^* = - \left(E - \frac{dE}{dq} q \right) \quad (18.31)$$

Note that only in the *sketch* of the curve $e = q/C$ a linear relation is assumed. Due to this it holds only *in this case* that $E = E^*$. This does not hold in the generic, non-linear case. Equation (18.31) holds in all cases, though. The term between brackets in (18.31) is also called the *Legendre transform* or the *Legendre–Fechel transform*.

Legendre-
transform

$$\mathcal{L}\{E(q)\} = E - \frac{dE}{dq} q \quad (18.32)$$

See H.B. Callen, *Thermodynamics*, Wiley, 1960, pp 90–98, for the use of the Legendre transform in thermodynamics and G. Strang, *Introduction to Applied Mathematics*, Wellesley–Cambridge Press, 1986, pp. 253 and 731, for a formal mathematical treatment of the Legendre-transform.

Combination of (18.31) and (18.32) results in:

$$E^* = -\mathcal{L}\{E(q)\} \quad (18.33)$$

or, using $E^* = E^*(e)$

$$E^*(e) = -\mathcal{L}\{E(q)\} \quad (18.34)$$

The co-energy $E^*(e)$ thus equals the negative value of the Legendre transform of $E(q)$ with respect to q .

Note that many treatments exist in which the co-energy is confused with the Legendre transform itself, thus ignoring the minus sign. However, Figure 18.5 clearly demonstrates that the sum of energy and complementary energy or co-energy should result in the product of intensive state or effort e and extensive state q , $E^* = eq - E$ or $E + E^* = eq$ in contrast with the Legendre transform

$$\mathcal{L}\{E(q)\} = E - \frac{dE}{dq} q = E - eq \quad \text{or} \quad E - \mathcal{L}\{E(q)\} = eq.$$

This Legendre transform can be extended to functions of more than one variable, e.g. the energy function characterizing a multiport storage element. This is postponed to section 7.

3.2 GIBBS' RELATION

Internal energy

The previous chapter already mentioned that Gibbs was the first to formulate the first law of thermodynamics in terms of energy relations. In thermodynamics this energy is often restricted to what is called the *internal energy* of an object, i.e. the energy directly related to the interactions between the matter that is contained in the system. This excludes for example the kinetic energy with respect to a point of body located outside the object under consideration or the potential energy of that body in an external force field (for example the gravitational force field), because these phenomena are considered to belong to the domain of mechanics. In this way an undeserved (cultural) separation has emerged between mechanics —where both q -type and p -type variables are used— and thermodynamics — where only q -type variables are used— while the energy conservation principle holds for *all* domains.

An often-used argument in thermodynamics is that the changes of state of a storage element need to be quasi-static, such that the kinetic cannot play a role. However, it is neglected that a similar argument holds in mechanics. Not only changes in potential energy need to take place quasi-statically in order to be representable by a storage element relation, but also changes in kinetic energy, although in this case the terminology quasi-stationary changes of the momentum would be more adequate. In other words: Newton's second law only holds for quasi-stationary changes of the momentum. This does *not* mean that it is assumed that *all* changes of state are quasi-stationary. However, it *does* mean that these changes can be separated into a quasi-stationary part and a non-quasi-stationary part that needs to contribute to some form of entropy production by definition.

Example

As shown in chapter 5, an ideal mass-spring system has a sinusoidal solution, i.e. it performs a permanent oscillation. However, when its frequency is further increased, the system will 'radiate' acoustically, in other words: the oscillation becomes audible. This means that the mass-spring system loses energy in the form of radiated acoustic energy, unless the oscillator is contained in vacuum. In this radiating situation the condition of quasi-stationary changes of state is not satisfied anymore.

Just like electric circuits are assumed not to be radiating electromagnetic energy and thus to be quasi-stationary, this is also assumed for mechanical systems, but this does not mean that kinetic energy has been set offside, like many thermodynamic studies suggest.

In thermodynamics only q -type storage elements are used. In the most other domains p -type storage elements can be found. For a q -type storage element holds that

$$dE = \frac{dE}{dq} dq \quad (18.35)$$

or, in case of an n -port storage element

$$dE(\mathbf{q}) = \sum_{i=1}^n \frac{\partial E(\mathbf{q})}{\partial q_i} dq_i \quad (18.36)$$

where $\mathbf{q}$ is an n -dimensional column matrix ('vector'). If a p -type storage element is considered this can be written

$$dE(\mathbf{p}) = \sum_{i=1}^n \frac{\partial E(\mathbf{p})}{\partial p_i} dp_i \quad (18.37)$$

However, one can also use the convention that only (18.36) is used. In that case the total differential dE becomes for *all* storage elements and for *all* domains:

$$dE(\mathbf{q}) = \sum_{i=1}^n \frac{\partial E(\mathbf{q})}{\partial q_i} dq_i \quad (18.38)$$

where $\mathbf{q}$ an n -dimensional column matrix ('vector') with both $\mathbf{q}$ - and $\mathbf{p}$ -type variables.

Furthermore, it holds that

$$e_i = \frac{\partial E(\mathbf{q})}{\partial q_i} \quad (18.39)$$

Substitution in (18.38) gives

$$dE(\mathbf{q}) = \sum_{i=1}^n e_i dq_i \quad (18.40)$$

Gibbs' fundamental relation

Generalized effort / state

This is a generalized form of the *Gibbs' fundamental relation*.

Note that e_i is an *effort* when q_i is of the q -type and that e_i is a *flow* when q_i is of the p -type. This explanation will not be repeated but the terminology *generalized effort* e_i and *generalized state* q_i will be used to indicate this.

Using a column matrix notation $\mathbf{e}$ for the generalized efforts a more compact form is obtained:

$$dE = \mathbf{e}^T d\mathbf{q} \quad (18.41)$$

Generalized flow

If a *generalized flow* f_i is introduced too according to

$$f_i = \frac{dq_i}{dt} = \dot{q}_i \quad (18.42)$$

and

$$\mathbf{f} = \frac{d\mathbf{q}}{dt} \quad (18.43)$$

then the total power supplied to a storage element can be written

$$P = \frac{dE}{dt} = \sum_{i=1}^n e_i f_i = \mathbf{e}^T \mathbf{f} \quad (18.44)$$

This kind of storage can be represented in a bond graph by a C–element with n ports (bonds). The previous chapter already showed that such a multiport storage element is often represented by a ‘shaded’ letter C in order to be able to distinguish it from the one–port storage element (figure 18.6).

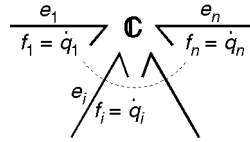


FIGURE 18.6
Multiport C with n bonds

The n bonds representing the ports can be noted more compact by means of a so-called multibond (figure 18.7 and figure 18.8)

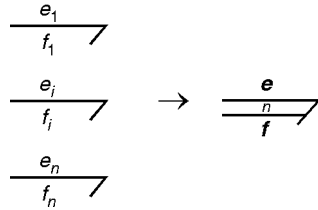


FIGURE 18.7
 n ports noted as multibond with dimension n

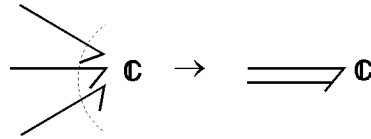


FIGURE 18.8
Multiport C with multibond

Just like a one–port storage element a multiport storage element is characterized by constitutive relations. It was shown earlier that one constitutive relation per port is needed, such that an n –port storage element requires n constitutive relations of the form

$$e_i = e_i(\mathbf{q}) \quad \text{for all } i: 1 \leq i \leq n \quad (18.45)$$

for which holds that

$$de_i = \sum_{j=1}^n \frac{\partial e_i(\mathbf{q})}{\partial q_j} dq_j \quad \forall i: 1 \leq i \leq n \quad (18.46)$$

or in matrix notation:

$$d\mathbf{e} = \frac{\partial \mathbf{e}_{i,j}(\mathbf{q})}{\partial q_{i,j}} d\mathbf{q} \quad (18.47)$$

If these relations are linear in all q_i 's they can be written

$$e_i = \sum_{j=1}^n \frac{\partial e_i(\mathbf{q})}{\partial q_j} q_j = \sum_{j=1}^n a_{ij} q_j \quad \text{for all } i: 1 \leq i \leq n \quad (18.48)$$

where $a_{ij} = \partial e_i(\mathbf{q})/\partial q_j = \text{constant}$ by definition, such that

$$\mathbf{e} = \frac{\partial e_{i,j}(\mathbf{q})}{\partial q_{i,j}} \mathbf{q} = \mathbf{A}\mathbf{q} = \mathbf{C}^{-1}\mathbf{q} \quad (18.49)$$

where $\mathbf{A}$ is the matrix with matrix elements a_{ij} and $\mathbf{C}$ is called the capacitance matrix. However, these constitutive relations are commonly non-linear.

ASSIGNMENT 18.14

Show that the constitutive relations of a capacitor with a moving plate, a water container with float and an elastic coil are non-linear.

Example

An example of a model with a multiport $\mathbf{C}$ that is linear is formed by two coupled coils, like in an electric transformer. The flux linkage of these coils is determined by the current through the coil itself ($\lambda = Li_1$), but also by the current in the other coil ($\lambda = Mi_2$), such that:

$$\lambda_1 = L_1 i_1 + M i_2 \quad (18.50)$$

and

$$\lambda_2 = M i_1 + L_2 i_2 \quad (18.51)$$

or in matrix form

$$\begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix} = \begin{pmatrix} L_1 & M \\ M & L_2 \end{pmatrix} \begin{pmatrix} i_1 \\ i_2 \end{pmatrix} \quad (18.52)$$

ASSIGNMENT 18.15

Does the matrix form of (18.52) correspond to the preferred causality? When yes, list the generalized q -variables and the generalized e -variables. When no, derive the form that corresponds to the preferred causality.

4 Maxwell reciprocity

Note that the matrix in the relation (18.52) is again symmetric. This observation was also made in case of the non-linear examples of the flat-plate capacitor, et cetera, where the linear differential form also contains a symmetric matrix (cf. (18.9), (18.15) and (18.23)). This property can now be analyzed from another perspective, because it is now clear that $a_{ij} = a_{ji}$ means that $\partial e_i(\mathbf{q})/\partial q_j = \partial e_j(\mathbf{q})/\partial q_i$ and consequently that —cf. the definition of $e_j(\mathbf{q})$ —

$$\frac{\partial^2 E(\mathbf{q})}{\partial q_i \partial q_j} = \frac{\partial^2 E(\mathbf{q})}{\partial q_j \partial q_i} \quad \text{for all } (i,j): 1 \leq i,j \leq n \quad (18.53)$$

In case $i = j$ this is a trivial statement. For $i \neq j$ it means that the mixed second partial derivatives of the energy with respect to the stored states need to be equal. It can be shown that when this would not be the case, E would not be a conserved quantity. The formal proof goes beyond the frame of this text, but an

Maxwell-
reciprocity

example will demonstrate later that energy can be generated when this equality is not satisfied. The relations (18.53) describing in the linear case the symmetry of the constitutive matrix —the capacitance matrix $\mathbf{C}$ — are called the *Maxwell reciprocity relations*, named after the already known *James Clerk Maxwell* who formulated them first. They are also called the *Maxwell reciprocity* of the multiport storage element. The earlier suspicion that the symmetry is no coincidence appears to be justified: it is a direct consequence of the first law.

ASSIGNMENT 18.16

Show that the constitutive relations (17.66) and (17.67) for a constant amount of ideal gas (N moles) satisfy the Maxwell reciprocity condition.

5 Intrinsic stability

Positive definite

In order to guarantee intrinsic stability of a one–port storage element it already appeared that its capacitance needs to be positive. In case of a multiport storage element this condition can be generalized to the *positive definiteness* of the capacitance matrix. A matrix $\mathbf{C} = [c_{ij}]$ is positive definite if the determinant $|\mathbf{C}| \neq 0$ and if the *quadratic form* of the matrix satisfies the following inequality:

$$\mathbf{x}^T \mathbf{C} \mathbf{x} > 0 \quad \text{for all } \mathbf{x} \neq \mathbf{0} \quad (18.54)$$

In case of a symmetric matrix, like the capacitance matrix, the following inequalities for the matrix elements can be derived:

$$c_{ii} > 0 \quad (18.55)$$

$$\frac{c_{ii}c_{jj} - c_{ij}^2}{c_{jj}} > 0 \quad i \neq j \quad (18.56)$$

for all (i,j) . In case of a two–port these inequalities mean that the determinant should be positive.

$$c_{11}c_{22} - c_{12}^2 > 0 \quad (18.57)$$

Criteria for
intrinsic stability
Coupling factor

Within the context of this text the validity of these *criteria for intrinsic stability* are only used, not proven. For a two–port storage element a *coupling factor* k is defined. The coupling factor is the square root of the product of the (equal) non–diagonal elements divided by the product of the diagonal elements

$$k = \left[\frac{a_{12}^2}{a_{11}a_{22}} \right]^{0.5} = \left[\frac{c_{12}^2}{c_{11}c_{22}} \right]^{0.5} \quad (18.58)$$

where a_{ij} and c_{ij} are the elements of the matrices $\mathbf{A}$ and $\mathbf{C}$ in (18.49). Based on (18.57) it holds for the coupling factor of an (intrinsically) stable two–port that $0 < k < 1$. The coupling factor can be interpreted as a relative measurer for the coupling of the ports of a two–port storage element via the off–diagonal terms with respect to the ‘self coupling’ of the both ports, the diagonal terms. Its both extremes, 0 and 1, correspond to complete decoupling (being separated) and full

coupling respectively. In the latter case one of the two states becomes fully dependent on the other state.

ASSIGNMENT 18.17

Compute the coupling factors of the capacitor with moving plate, the elastic coil and the water container with float.

*Principle of Le
Châtelier(–Braun)*

In thermodynamics these two criteria for intrinsic stability are known as the *principle of Le Châtelier* that can be formulated: *during a disturbance from the equilibrium state spontaneous processes will occur that drive the system back to equilibrium* and the *principle of Le Châtelier–Braun* that adds to this that *also all processes indirectly caused by the disturbance will drive the system back to equilibrium*.

The French chemist *Henry–Louis Le Châtelier* (1850–1936) derived this principle in 1884 in order to be able to predict the influence of variations in pressure, temperature and concentration on chemical reactions. Partially his work was an ignorant copy of the work by Gibbs.

ASSIGNMENT 18.18

Compute the coupling factor of two coupled coils and express the intrinsic stability condition in the properties of L_1 , L_2 and M .

The cause of the symmetry is found (Maxwell reciprocity) and the intrinsic stability condition is extended to the case of a multiport. However, we did not explain yet how, for example, a change of the geometric properties that results in a change of variables described by the examples, can be properly handled in an energy consideration.

ASSIGNMENT 18.19

How can the bond graphs found in assignment 18.10 be written after interchanging the dependent and the independent variables as described for a flat–plate capacitor, a water container with float and an elastic coil, i.e. instead of the independent variables q , V and λ , now u , p and i respectively are used as the independent variables? Also show what happens to these bond graphs when $k = 0$ or $k = 1$.

In the next sections the formal mathematical basis of these causality changes is considered. This requires the introduction of the concept *homogenous function* first.

6 Homogenous functions and the Euler equation

A careful consideration of energy functions prevents the sign problems encountered in the earlier examples. This section first shows the properties of each energy function. In the meantime it was shown that the generalized efforts and consequently the constitutive relations of a multiport in integral causality could be fully derived from the energy function by partial differentiation with respect to the stored quantities. This energy is in case of a multiport storage element a function of more than one variable. In fact these operations assumed that this function is a *homogenous function*. For a homogenous function $F(\mathbf{x})$ it holds by definition that

$$F(a\mathbf{x}) = a^n F(\mathbf{x}) \quad (18.59)$$

*Homogenous
function*

In this section F is an arbitrary function and not a force.

*Degree of the
homogenous
function*

where n is the *degree of the homogenous function*.

Example

An example of a homogenous function is:

$$F(\mathbf{x}) = F(x, y, z) = 2xyz^2 + 3x^2y^2 + x^3z \quad \text{where } \mathbf{x} = x, y, z$$

As each term is of the fourth power this a homogenous function of the fourth degree. After all:

$$\begin{aligned} F(a\mathbf{x}) &= F(ax, ay, az) = 2axaya^2z^2 + 3a^2x^2a^2y^2 + a^3x^3az \\ &= a^4(2xyz^2 + 3x^2y^2 + x^3z) \end{aligned}$$

Euler's theorem

In fact also linearity was used during the discussion of the homogeneity demand, which is limited to the first degree. For n -th degree homogenous functions *Euler's theorem* holds:

$$\sum_{i=1}^k \frac{\partial F(\mathbf{x})}{\partial x_i} x_i = nF(\mathbf{x}) \quad (18.60)$$

where $\mathbf{x}$ consists of k elements.

This theorem is named after the Swiss mathematician and physicist *Leonard Euler* (1707–1783), who is considered one of the founders of pure mathematics.

Suppose that

$$\frac{\partial F(\mathbf{x})}{\partial x_i} = y_i(\mathbf{x}) \quad (18.61)$$

then it follows from (18.60)

$$F(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^k y_i x_i = \frac{1}{n} \mathbf{y}^T \mathbf{x} \quad (18.62)$$

For $y_i(\mathbf{x})$ holds

$$y_i(a\mathbf{x}) = \frac{\partial F(a\mathbf{x})}{\partial ax_i} = \frac{a^n \partial F(\mathbf{x})}{a \partial x_i} = \frac{a^{n-1} \partial F(\mathbf{x})}{\partial x_i} = a^{n-1} y_i(\mathbf{x}) \quad (18.63)$$

which means that $y_i(\mathbf{x})$ is a homogenous function of degree $(n-1)$ according to the definition.

ASSIGNMENT 18.20

Show that Euler's theorem holds for the function:

$$F(\mathbf{x}) = F(x, y, z) = 2xyz^2 + 3x^2y^2 + x^3z \quad \text{where } \mathbf{x} = x, y, z$$

ASSIGNMENT 18.21

Suppose the constitutive relations of a multiport storage element are linear. Are these relations homogenous functions? If yes, what is the degree and what is the degree of the corresponding energy function? If no, give the additional conditions to obtain homogeneity.

The total differential $dF(\mathbf{x})$ of $F(\mathbf{x})$ equals

$$dF(\mathbf{x}) = \sum_{i=1}^k \frac{\partial F}{\partial x_i} dx_i \quad (18.64)$$

By means of (18.61) this can be written:

$$dF(\mathbf{x}) = \mathbf{y}^T d\mathbf{x} \quad (18.65)$$

From (18.62) follows:

$$dF(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^k d(y_i x_i) = \frac{1}{n} \mathbf{y}^T d\mathbf{x} + \frac{1}{n} d\mathbf{y}^T \mathbf{x} \quad (18.66)$$

such that

$$d\mathbf{y}^T \mathbf{x} = (n-1) \mathbf{y}^T d\mathbf{x} \quad (18.67)$$

This relation is now translated into a homogenous energy function $E(\mathbf{q})$:

$$dE(\mathbf{q}) = \mathbf{e}^T d\mathbf{q}$$

This can be written

$$dE(\mathbf{q}) = \frac{1}{n} \sum_{i=1}^k d(e_i q_i) = \frac{1}{n} \mathbf{e}^T d\mathbf{q} + \frac{1}{n} (d\mathbf{e})^T \mathbf{q} \quad (18.68)$$

such that

$$(d\mathbf{e})^T \mathbf{q} = (n-1) \mathbf{e}^T d\mathbf{q} \quad (18.69)$$

Homogenous systems with storage are systems in which no effort differences occur.

As homogenous systems with storage can always be divided by means of conceptual boundaries into a number of a homogenous storage elements without changing the total stored energy, it should hold in principle that

$$E(a\mathbf{q}) = aE(\mathbf{q}) \quad (18.70)$$

In other words: the energy should be a first-degree homogenous function of all stored quantities in principle. This also means that constitutive relations are always zero degree homogenous functions in principle, i.e.

$$\mathbf{e}(a\mathbf{q}) = \mathbf{e}(\mathbf{q}) \quad (18.71)$$

Example

The capacitor in figure 18.9 is used as an example. This capacitor, with a capacitance C and a charge q can be split into three other capacitors of equal capacitance, each charged with $q/3$ and each with a capacitance $C/3$.

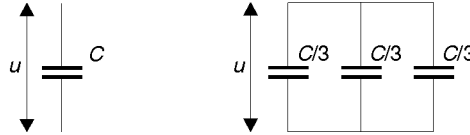


FIGURE 18.9 Separation of a capacitor in 3 capacitors

Obviously, the effort (voltage) remains the constant during this separation. For the stored energy it holds that:

$$E = \int u dq = \frac{1}{2C} q^2 = 3 \frac{3}{2C} \left(\frac{q}{3} \right)^2$$

*Intensive and
extensive
variables*

Extensive originates
from Latin: extendere
= to extend.

Consequently the generalized q -type variables need to be added during addition of homogenous subsystems, while the conjugated efforts remain the same. Based on the proportionality of the storage element contents with the number(s) of homogenous subsystems, the *storage element contents* are also called *extensive variables*. By contrast, the corresponding generalized *efforts*, which remain constant in the subsystems that are in equilibrium, are called *intensive variables*.

The above means among other things that in case of a one-port storage element the (generalized) effort would always remain constant, which contradicts the idea of a storage element. This paradox leads to the conclusion that storage elements are always multiports in principle. The latter important conclusion can be derived from

$$de^T q = (n-1)e^T dq \quad (18.72)$$

such that for $n = 1$ according to (18.72):

$$de^T q = 0 \quad (18.73)$$

*Gibbs–Duhem
relation*

This relation, called the *Gibbs–Duhem relation* in thermodynamics appears domain independent and will be called generalized Gibbs–Duhem equation herein.

Pierre Maurice Marie Duhem (1861–1916) was a French mathematician, physicist and etymologist.

Equation (18.73) can be considered as a *weighted sum* of generalized effort variations de_i , with *weighting* factors q_i that is zero, i.e. one of the efforts depends on the other efforts:

$$e_n = e_n(e_1, \dots, e_{n-1}) \quad (18.74)$$

These conclusions might suggest that impossible concepts have been used until now. However, the one-port storage elements used so far were in the above

context not one-port storage elements, but could be treated that way. If a state remains constant, then the flow of the corresponding storage element port is zero and thus its corresponding power. Omitting the port in that case has no influence on the dynamic behavior. Hence, a one-port storage element can be considered a storage element of which all states except one are constant. If a state constant remains constant, its conjugate effort is not of interest, as this effort is a function of the other efforts and consequently not needed for the description of the system.

Example

An electric capacitor is an example of this situation. The possibility of a mechanical port next to the electric port has been demonstrated already with the example of the capacitor with moving plate. However, if the distance between the plates is constant, then the mechanical port can be omitted and a one-port storage element remains.

In almost all thermodynamic systems treated until now the flow of the material port, of which the total material potential μ_{tot} is a function of the other efforts, is kept zero. This keeps the amount of matter in the storage element constant.

Another important conclusion can be drawn from (18.62).

For $F = E$, $\mathbf{x} = \mathbf{q}$, $\mathbf{y} = \mathbf{e}$ with $n = 1$:

$$E(\mathbf{q}) = \frac{1}{n} \sum_{i=1}^k e_i q_i = \sum_{i=1}^k e_i q_i = \mathbf{e}^T \mathbf{q} \quad (18.75)$$

Based on the definition of the concept *internal energy* U in thermodynamics, the internal energy $U(S, N, V)$ is a homogenous function of the *first* degree in the variables S (entropy), N (number of moles) and V (available volume). An amount of matter that is a times as large should contain an amount of internal energy that is a ($=a^1$) times as large. In this case (18.75) takes the form

$$U = -pV + \mu_{\text{tot}}N + TS \quad (18.76)$$

with the following definitions for the pressure p : $\partial U / \partial V = -p$, the total material potential μ_{tot} : $\partial U / \partial N = \mu_{\text{tot}}$, and the temperature T : $\partial U / \partial S = T$. This specific form of (18.75) is called the *Euler equation*. In thermodynamics the Euler equation is usually written in its intensive form, i.e. per mole of matter:

$$\begin{aligned} u(s, v) &= U\left(\frac{S}{N}, 1, \frac{V}{N}\right) = \left(\frac{1}{N}\right)^1 U(S, N, V) = \\ &= T \frac{S}{N} + \mu_{\text{tot}} \frac{N}{N} - p \frac{V}{N} = Ts + \mu_{\text{tot}} - pv \end{aligned} \quad (18.77)$$

so

$$u(s, v) = Ts + \mu_{\text{tot}} - pv \quad (18.78)$$

where by definition $u(s, v) = U(S, 1, V)$, $s = S/N$ and $v = V/N$. Note that $u(s, v)$ is not a first degree homogenous function in s and v . This intensive form of the Euler

equation can be used to find an expression for μ_{tot} in this simple thermodynamic case:

$$\mu_{\text{tot}} = -Ts + pv + u(s, v) \quad (18.79)$$

For an ideal gas it holds that

$$u(s, v) = c_v T \quad (18.80)$$

and

$$pv = RT = (c_p - c_v)T \quad (18.81)$$

such that in this case holds for μ_{tot} :

$$\mu_{\text{tot}} = -Ts + RT + c_v T = T(c_p - s) \quad (18.82)$$

Generalized Euler equation

Based on the name used in thermodynamics (18.75) is called herein a *generalized Euler equation*. Outside of thermodynamics it is not often encountered, because the amount of matter is mostly kept constant ($dN = 0$). This means that a variable, the amount of matter, can be omitted from the energy function. Consequently the degree n of the homogenous energy function increases to $n = 2$. In case the remaining constitutive relations are linear, (18.62) becomes common form for linear mechanical and electric systems

$$E(\mathbf{q}) = \frac{1}{2} \mathbf{e}^T \mathbf{q} \quad (18.83)$$

This generalized Euler equation is important for the systematic and energetically correct description of convection processes of all sorts of properties other than entropy, for example momentum, charge or magnetic flux. However, as noted before, its elaboration is beyond this context.

7 Legendre transforms

Section 3 showed that if a storing port does not have integral causality, then co-energy instead of energy should be computed. It was shown for a (energy) function of one variable that the co-energy can be obtained by means of a Legendre transform. In this section this transformation is extended to (energy) functions of more than one variable. The consequences of the exchange of q_i and e_i for an energy function $E(q)$ are studied. In other words: the exchange of x_i and y_i for an arbitrary homogenous function $F(x)$ are to be analyzed.

Legendre transform

The *Legendre transform* of $F(x)$ with respect to the variable x_i —for short

$\mathcal{L}\{F(\mathbf{x})\}|_{x_i}$ — is defined

$$\mathcal{L}\{F(\mathbf{x})\}|_{x_i} = F(\mathbf{x}) - y_i x_i \quad \text{where } y_i = \frac{\partial F(\mathbf{x})}{\partial x_i} \quad (18.84a)$$

or, in terms of an energy function:

$$\mathcal{L}\{E(\mathbf{q})\}\Big|_{q_i} = E(\mathbf{q}) - e_i q_i \quad \text{with } e_i = \frac{\partial E(\mathbf{q})}{\partial q_i} \quad (18.84b)$$

This means for the total differential of this Legendre transform that

$$\begin{aligned} d\mathcal{L}(F)\Big|_{x_i} &= dF(\mathbf{x}) - y_i dx_i - x_i dy_i = \\ &= \mathbf{y}^T d\mathbf{x} - y_i dx_i - x_i dy_i \\ &= y_1 dx_1 + y_2 dx_2 + \dots - x_i dy_i + \dots + y_n dx_n \\ &= \sum_{j \neq i}^n (y_j dx_j) - x_i dy_i \end{aligned} \quad (18.85)$$

The corresponding Legendre transform can be written in generic form:

$$\mathcal{L}\{F(x_1, \dots, x_{i-1}, x_i, x_{i+1}, \dots, x_n)\}\Big|_{x_i} = F(x_1, \dots, x_{i-1}, y_i, x_{i+1}, \dots, x_n) \quad (18.86a)$$

or, in terms of an energy function:

$$\mathcal{L}\{E(q_1, \dots, q_{i-1}, q_i, q_{i+1}, \dots, q_n)\}\Big|_{q_i} = -E(q_1, \dots, q_{i-1}, e_i, q_{i+1}, \dots, q_n) \quad (18.86b)$$

where $(x_1, \dots, x_{i-1}, y_i, x_{i+1}, \dots, x_n)$ represent the arguments of the resulting function. This transform replaces x_i (or q_i) by y_i (e_i) as the independent variable. The importance of this should not be underestimated. Often the exchange of y_i and x_i is not treated with sufficient care, especially when e_i and q_i in energy functions are concerned.

The energy, for which a conservation principle holds, is a function of the storage element contents, i.e. the generalized q -variables. Replacing one of the q -variables by its conjugated effort means a *Legendre transform* with respect to q_i . However, note well that a Legendre transform of an energy is *not a conserved quantity* anymore. A previously encountered Legendre transform, albeit implicitly, is the ‘free energy’ $\mathcal{H}(\mathbf{q}, T)$, a Legendre transform of the energy with respect to the entropy S (figure 18.10).

Example

Consider the following energy function:

$$E(\mathbf{q}) = E(q_1, q_2, \dots, q_{n-1}, q_n)$$

Suppose $q_n = S$, where S is the entropy belonging to port n . If this port should obtain flow-out causality, the effort (T) should become the independent variable. To this end a Legendre transform can be performed with respect to q_n :

$$\mathcal{L}\{E(\mathbf{q})\}\Big|_{q_n=S} = E(\mathbf{q}) - TS = F(\mathbf{q}', T) \quad \text{where } \mathbf{q}' = (q_1, q_2, \dots, q_{n-1})$$

The dimension of the $\mathbf{q}'$ -vector and thus of the left port in figure 18.10 becomes $n-1$, because S is not part of the vector $\mathbf{q}'$ anymore. Figure 18.10 shows how this can be represented in a bond graph.

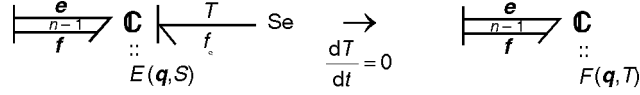


FIGURE 18.10 Free energy

In other words: if it is assumed that the temperature of the environment remains constant, such that the thermal domain is not considered, free energy can be used. This was already done in the first two parts of this text, but now it has been justified.

The free energy is usually indicated by the symbol $\mathcal{F}$ for *Free energy*. In order to prevent confusion with F as a symbol for a function, $\mathcal{F}$ is used for the free energy.

(Helmholtz) free energy

The free energy $\mathcal{F}$ was first used in 1882 by the German physicist *Hermann Ludwig Ferdinand von Helmholtz* (1821–1894) and is also called the *Helmholtz free energy* accordingly.

Another symbol that is used is $A(T, \mathbf{q})$. It holds for the free energy $\mathcal{F}(=A)$:

$$\mathcal{F}(\mathbf{q}, T) = \mathcal{L} \{ E(\mathbf{q}, S) \} \Big|_S = E(\mathbf{q}, S) - TS \quad (18.87)$$

and

$$d\mathcal{F}(\mathbf{q}, T) = dE(\mathbf{q}, S) - TdS - SdT = (dE(\mathbf{q}, S) - TdS) - SdT \quad (18.88)$$

Cf. figure 18.10.

The term $-TdS$ represents the changes of free energy through the non-thermal ports (with integral causality). The term $-SdT$ represents the changes of free energy through the thermal port. In the most cases is the reason for the differential causality of this port is a temperature source that is constant (ambient temperature). In that case $dT = 0$, such that

$$\begin{aligned} d\mathcal{F}(\mathbf{q}) \Big|_{dT=0} &= dE(\mathbf{q}, S) - TdS \\ &= \left(\sum_{i=1}^{n-1} e_i dq_i + TdS \right) - TdS \\ &= \sum_{i=1}^{n-1} e_i dq_i = \mathbf{e}^T d\mathbf{q} \end{aligned} \quad (18.89)$$

Note well that S is not part of the $\mathbf{q}$ -vector and consequently that T is not part of the $\mathbf{e}$ -vector. In thermodynamics two more Legendre transforms have a separate name apart from the free energy $\mathcal{F}$:

– Enthalpy:

If the ambient pressure is constant it is useful to work with a Legendre transform of the energy $E(V, \mathbf{q})$ with respect to the volume. This quantity is

Enthalpy

The Greek word *θαλπος* ('thalpos') means 'heat'; the prefix *εν-* ('en-') means 'in'.

called *enthalpy* $H(p, q)$:

$$H = E - (-p)V = E + pV \quad (18.90)$$

$$dH = \mathbf{e}^T dq + Vdp \quad (18.91)$$

$$dH|_{dp=0} = \mathbf{e}^T dq \quad (18.92)$$

– Gibbs' free energy:

Gibbs' free energy
Gibbs' free enthalpy

If both pressure and temperature are constant, the Legendre transform of the energy $E(V, S, q)$ with respect to the volume as well as the entropy is often used. This is called the *Gibbs' free energy* $G(p, T, q)$. The name *Gibbs' free enthalpy* is sometimes used too.

$$G = E - (-p)V - TS = E + pV - TS = H - TS = F + pV \quad (18.93)$$

$$dG = \mathbf{e}^T dq + Vdp - SdT \quad (18.94)$$

$$dG|_{dp=dT=0} = \mathbf{e}^T dq \quad (18.95)$$

Note that q represents another column matrix in these cases, viz. 'the remaining states', that are not involved in the transformation. In fact this form is kept more generic than usual, because in thermodynamics the restriction to the internal energy U is made during the discussion of the above Legendre transforms.

Legendre transforms are domain independent: energy of an electric circuit expressed in coil currents and capacitor voltages is a Legendre transform of the energy stored in the circuit with respect to all charges and flux linkages. The kinetic 'energy' expressed in the velocity v (the common $\frac{1}{2}mv^2$) is in fact a Legendre transform of the true kinetic energy with respect to the momentum p and in itself not a true energy anymore, et cetera.

ASSIGNMENT 18.22

Show that the kinetic 'energy', expressed in the form $\frac{1}{2}mv^2$, is a co-energy.

The question immediately rises how it has been possible that so far (in line with common treatments in the literature) e and q could be exchanged 'scot-free' during the computation of the energy? The answer is both simple and disturbing: there was a constraint to linear constitutive relations. First it is shown why this has a correcting effect. In case of linear constitutive relations it holds that the degree (of the linear function) equals 1: $n - 1 = 1$, so $n = 2$, where n is the degree of the homogenous energy function. This means that (cf. (18.69)):

$$(\mathbf{de})^T q = (n-1)\mathbf{e}^T dq = \mathbf{e}^T dq \quad (18.96)$$

and (cf. (18.68))

$$\begin{aligned}
 dE(\mathbf{q}) &= \frac{1}{n} \mathbf{e}^T d\mathbf{q} + \frac{1}{n} (d\mathbf{e})^T \mathbf{q} = \\
 &= \frac{1}{2} \mathbf{e}^T d\mathbf{q} + \frac{1}{2} (d\mathbf{e})^T \mathbf{q} = d\mathbf{e}^T \mathbf{q} = \mathbf{e}^T d\mathbf{q}
 \end{aligned} \tag{18.97}$$

Consequently, the exchange of $\mathbf{e}$ and $\mathbf{q}$ seems to have no effect.

Total
Legendre-
transform

The *total Legendre transform* is a transform with respect to all x_i 's:

$$\mathcal{L}\{E(\mathbf{q})\}\bigg|_{\mathbf{q}(\mathbf{e})} = E(\mathbf{q}) - \mathbf{e}^T \mathbf{q} = E(\mathbf{q}) - nE(\mathbf{q}) = (1-n)E(\mathbf{q}) \tag{18.98}$$

For the case where $n = 1$ the *total Legendre transform* is zero.

$$\mathcal{L}\{E(\mathbf{q})\}\bigg|_{\mathbf{q}(\mathbf{e})} = \mathcal{L}(\mathbf{e}) = 0 \tag{18.99}$$

This confirms the conclusion that one of the generalized efforts depends on the other ones. (Cf. (18.74).)

$$e_i = e_i(e_1, e_2, \dots, e_{i-1}, e_{i+1}, \dots, e_k) \tag{18.100}$$

For the linear case ($n = 2$) it holds that

$$\mathcal{L}\{E(\mathbf{q})\}\bigg|_{\mathbf{q}(\mathbf{e})} = \mathcal{L}(\mathbf{e}) = -E(\mathbf{q}) \tag{18.101}$$

Co-energy or
complementary
energy

The sign inverse of the (total) Legendre transform is also called the *co-energy* or *complementary energy* $E^*(\mathbf{e})$, already encountered in part 1 and also in section 3 of this chapter.

$$E^*(\mathbf{e}) = -\mathcal{L}(\mathbf{e}) \tag{18.102}$$

According to (18.98):

$$E(\mathbf{q}) - \mathcal{L}(\mathbf{e}) = \mathbf{e}^T \mathbf{q} \tag{18.103}$$

Substitution of (18.102) results in:

$$E(\mathbf{q}) + E^*(\mathbf{e}) = \mathbf{e}^T \mathbf{q} \tag{18.104}$$

Compare this result with section 3.1 too. For $n = 2$ (cf. (18.62)):

$$E(\mathbf{q}) = \frac{1}{2} \mathbf{e}^T \mathbf{q} \tag{18.105}$$

From (18.104) and (18.105) follows:

$$E(\mathbf{q}) = E^*(\mathbf{e}) \tag{18.106}$$

such that the co-energy and the energy have equal magnitudes in the linear case. They are different in nature, as demonstrated by the examples of the flat-plate capacitor and the elastic coil.

ASSIGNMENT 18.23

Show how the force $F(u, x)$ at the plates of a flat-plate capacitor in terms of the voltage u and the distance between the plates x and the force $F(i, x)$ at the elastic coil in terms of the current i and the length x can be directly derived from the co-energy expressions.

The solution of the previous assignment also resolved the problem of the ‘missing’ minus sign in case of the capacitor and the elastic coil. It also becomes clear why more than a minus sign went wrong in case of the water container with float. The constitutive relation is non-linear (the p - V -characteristic has discontinuity in the derivative that depends on x). As a result energy and co-energy are not equal in value in that case.

ASSIGNMENT 18.24

Derive the force $F(p, x)$ at the float of the water container from the co-energy ($= -\mathcal{E}$).

The distinction between energy and co-energy is often considered a superfluous accuracy. The arguments shown so far should be sufficient to demonstrate this position is not correct.

The following text in small letters is background material.

*Hamiltonian
Lagrangian*

*Lagrange
equations*

In mechanics an amount of energy correctly formulated as a function of displacements $\mathbf{q}$ and momentum variables $\mathbf{p}$ is called Hamiltonian, named after the Irish mathematician and astronomer *Sir William Rowan Hamilton* (1805–1865). He was the first to realize that the expression of the mechanical energy in velocities and displacements leads to a Legendre transform of the energy. This Legendre transform of the Hamiltonian $H(\mathbf{q}, \mathbf{p})$ (not to be confused with the enthalpy, for which the letter H is also used) with respect to $\mathbf{p}$ is called, after sign inversion, the Lagrangian $L(\mathbf{q}, \mathbf{v})$, after the French–Italian mathematician Joseph Louis Lagrange (originally Giuseppe Luigi Lagrangia, 1736–1813). In his book ‘*Mécanique analytique*’ (1788) that largely influenced mechanics, he used this Lagrangian in order to derive the equations of motion via the so-called *Lagrange equations*. By means of the already introduced Legendre transforms these equations can be simply derived from Newton’s second law and from the definition of the velocity. First it is noted that the momentum $\mathbf{p}$ equals the partial derivative of the co-energy with respect to the velocity variables $\mathbf{v}$. In this case this co-energy is called the Lagrangian L :

$$\mathbf{p} = \frac{\partial H^*(\mathbf{q}, \mathbf{v})}{\partial \mathbf{v}} = \frac{\partial L(\mathbf{q}, \mathbf{v})}{\partial \mathbf{v}} \quad (18.107)$$

such that

$$\frac{d\mathbf{p}}{dt} = \frac{d}{dt} \frac{\partial L(\mathbf{q}, \mathbf{v})}{\partial \mathbf{v}} \quad (18.108)$$

According to Newton’s second law:

$$\frac{d\mathbf{p}}{dt} = \sum \mathbf{F} \quad (18.109)$$

If $\sum \mathbf{F}$ is split into forces $\sum \mathbf{F}_{\text{cons}}$ that can be derived from the potential energy by means of partial differentiation with respect to $\mathbf{q}$ (conservative forces) and consequently also by means of the negative partial derivative of the Lagrangian with respect to $\mathbf{q}$

$$\sum \mathbf{F}_{\text{cons}} = -\frac{\partial H^*(\mathbf{q}, \mathbf{v})}{\partial \mathbf{q}} = -\frac{\partial L(\mathbf{q}, \mathbf{v})}{\partial \mathbf{q}} \quad (18.110)$$

and forces $\sum \mathbf{F}_{\text{external}}$ that are not conservative and are also called ‘external forces’, such that —the minus sign corresponds to the common sign conventions for a bond graph, cf. figure 18.11—

$$\sum \mathbf{F} = \sum \mathbf{F}_{\text{external}} - \sum \mathbf{F}_{\text{cons}} \quad (18.111)$$

The minus sign of the term $\sum \mathbf{F}_{\text{cons}}$ is a consequence of the implicit sign convention in (18.110). The Lagrange equations result

$$\frac{d}{dt} \frac{\partial L(\mathbf{q}, \mathbf{v})}{\partial \mathbf{v}} - \frac{\partial L(\mathbf{q}, \mathbf{v})}{\partial \mathbf{q}} = \sum \mathbf{F}_{\text{external}} \quad (18.112)$$

If H is considered the sum of kinetic energy $T(\mathbf{q}, \mathbf{p})$ and potential energy $V(\mathbf{q})$:

$$H(\mathbf{q}, \mathbf{p}) = T(\mathbf{q}, \mathbf{p}) + V(\mathbf{q}) \quad (18.113)$$

its Legendre transform with respect to $\mathbf{p}$:

$$\begin{aligned} T(\mathbf{q}, \mathbf{p}) + V(\mathbf{q}) - \mathbf{v}\mathbf{p} &= T(\mathbf{q}, \mathbf{p}) + V(\mathbf{q}) - T(\mathbf{q}, \mathbf{p}) - T^*(\mathbf{q}, \mathbf{v}) = \\ &= V(\mathbf{q}) - T^*(\mathbf{q}, \mathbf{v}) \end{aligned} \quad (18.114)$$

where $T + T^* = \mathbf{v}\mathbf{p}$ (cf. (18.103)) is used, such that the Lagrangian $L(\mathbf{q}, \mathbf{v})$ equals

$$L(\mathbf{q}, \mathbf{v}) = -\left(V(\mathbf{q}) - T^*(\mathbf{q}, \mathbf{v})\right) = T^*(\mathbf{q}, \mathbf{v}) - V(\mathbf{q}) \quad (18.115)$$

Combination with the definition of the velocity $\mathbf{v} = d\mathbf{q}/dt$ results in

$$L\left(\mathbf{q}, \frac{d\mathbf{q}}{dt}\right) = T^*\left(\mathbf{q}, \frac{d\mathbf{q}}{dt}\right) - V(\mathbf{q}) = T^* - V \quad (18.116)$$

in other words: the Lagrangian is the difference of kinetic *co*-energy T^* and potential energy V , in contrast to the common formulation in the literature, where L is defined as $L = T - V$. The reason that this fundamental error commonly does not result in serious errors was demonstrated earlier in this chapter (cf. for example (18.104)): in the linear case the values of T and T^* are equal in value and the position dependent mass in mechanisms is not defined according to Newton’s second law.

Furthermore, it is recalled that Legendre transforms correspond to the change of the preferred causality into the differential causality. When the Lagrange equations are applied this occurs automatically, even without necessity. Hamilton derived the equations of motion too, but in the preferred causality:

$$\mathbf{v} = \frac{\partial H(\mathbf{q}, \mathbf{p})}{\partial \mathbf{p}} = \frac{d\mathbf{q}}{dt} \quad (18.117)$$

and

$$\sum \mathbf{F}_{\text{cons}} = \frac{\partial H(\mathbf{q}, \mathbf{p})}{\partial \mathbf{q}} = -\frac{d\mathbf{p}}{dt} + \sum \mathbf{F}_{\text{external}} \quad (18.118)$$

The combination of (18.117) and (18.118) describes a symplectic gyrator (figure 18.11). (Hamiltonian mechanics is said to have a symplectic structure.)

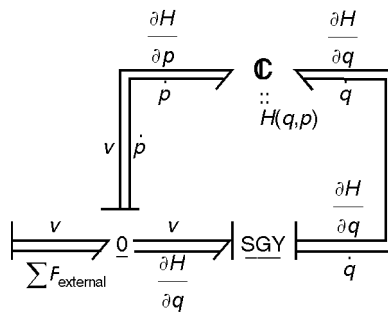


FIGURE 18.11
Bond graph of an equation of motion
according to Hamilton

Different examples were shown of the importance of expressing the energy in the correct independent variables, being the storage element contents (extensive variables). This makes the question whether physical criteria exist in order to distinguish these variables from the equilibrium determining variables. The answer lies in the observation of an addition process of storage elements where the total energy cannot change or cannot transfer to another domain. The consequence of this condition is that the equilibrium determining variables need to be equal, because, if no equilibrium would exist, irreversible processes would take place that transfer energy into the thermal domain. If the addition is performed, not only the energy of the newly obtained system is equal to the sum of the separate energies, but also the new storage element content is the sum of the separate, equivalent storage element contents.

Example

Combination of moving masses into one is only possible when they have equal velocities. The momenta do not have to be equal and the total momentum of the combination equals the sum of the individual momenta. Hence, the storage element content is the momentum, whereas the velocity can be considered the equilibrium-determining variable.

8 Summary

This chapter elaborated the difference between energy and co-energy and the relation of these quantities with the causality of a storage element. As long as linear systems are considered, the distinction seems superfluous. However, if the system is non-linear, as is the case for almost all realistic multiport storage elements, more care should be taken and the distinction between energy and co-energy cannot be neglected. In order to be able to perform these kinds of analyses correctly, also a number of more theoretical subjects should be considered. A generalized energy function was introduced that describes both q - and p -type storage ports. Furthermore, the concept of a homogenous function has been introduced.

After the treatment of the Legendre transform for multiport storage elements it became clear that co-energies have already been used unconsciously, like the kinetic co-energy $\frac{1}{2}mv^2$ that is commonly called 'kinetic energy' and the free energy.

The following chapter will apply this approach of multiport storage to describe energy transduction from one domain into the other. This kind of transduction requires a cycle process.

EXPERIMENTS

Perform the experiments in chapter 24 that correspond to chapter 18. If chapter 24 is not yet available to you, go to www.20sim.com, download the demo version of 20-sim, study the tutorial and try to use 20-sim for the simple problems of the current chapter where applicable.

SELF TEST

- 1 What is a Legendre transform?
- 2 Show that the energy stored in the flat-plate capacitor (figure 18.1) is a homogenous function. What is the degree of this function?
- 3 Are the following (energy) functions homogenous functions?

$$E(q) = a \exp(bq)$$

$$E(q_1, q_2) = a \exp\left(b \frac{q_1}{q_2}\right)$$

If yes, what is their degree?

- 4 When is it possible to express the energy of an electric circuit in the independent coil currents $\mathbf{i}$ and the independent capacitor voltages $\mathbf{u}$?
- 5 List practical examples of systems of which the dominant behavior can be described by a two-port storage element.
- 6 Construct a causal bond graph model of the dynamic behavior of a ship in a dock. The ship may be considered as an open container with vertical walls and a flat bottom. Derive the equations for the following two situations:
 - the dock is closed from the water in the harbor and is emptied by pumping the water from the bottom
 - the dock is in open connection with the water in the harbor.

F E E D B A C K

1 Solutions of the assignments

- 18.1 The lever should be infinitely stiff, otherwise it stores elastic energy during transduction. This condition is not realistic in the sense that infinitely rigid levers do not exist. However, within most problem contexts it is realistic to neglect the storing properties of the lever with respect to the dominant property, the transduction.
- 18.2 The electric transformer consisting of two coupled coils. This is a magnetic two-port storage element. Neglecting the storage properties enables transduction of DC-signals (DC = *direct current*), while in the practice only AC-signals (AC = *alternating current*) can be transformed by this device. The next chapter will demonstrate that AC-signals in fact maintain a cycle process.
- 18.3 The condition is that q_0 should not be zero. In case q_0 is zero the non-diagonal terms in the matrix are zero and there is no actual two-port, but just two separate one-ports.
- 18.4 Examples are the electrostatic microphone (electret microphone) and the electrostatic loudspeaker.
- 18.5 The horizontal line is only realistic when $x > 0$, because the plates cannot penetrate each other which would correspond with negative values of x . For $x = 0$ there is a galvanic contact between the plates such that no charge difference is possible. Accordingly, the characteristic is not defined for $x < 0$ and in case $x = 0$ it holds that $F = 0$. Furthermore, x cannot become so large that fringing effects disturb the linearity of the constitutive relation. If this would happen, then the force will decrease when x increases and finally become zero.
- 18.6 The force F is the force that needs to be applied to the plates in order to keep the plates at a distance x . This force is always positive. This means that one should pull in the positive direction (away from the other plate). In turn, this means that, if one would not pull, the plates would attract each other always and move towards each other until they touch. This phenomenon will be elaborated later.
- 18.7 If the case that the float is fully submerged is excluded (for example, by making the float longer than the height of the reservoir), two situations need to be distinguished:

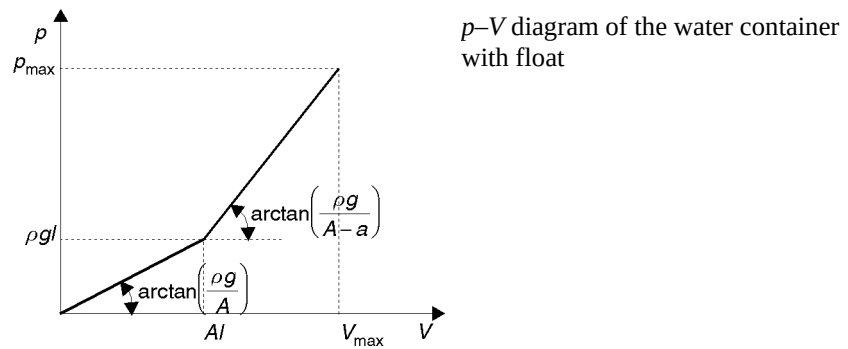
- a The case that the water level has not yet reached the float:

$$p = \frac{\rho g}{A} V$$

- b The case that the float is partially submerged:

$$p = \frac{\rho g}{A} Al + \left[\frac{\rho g}{A-a} \right] (V - Al) = \rho g \left[l + \frac{V - Al}{A-a} \right]$$

In the next figure this constitutive relation is sketched in a p - V diagram.



For the purpose of integration V is replaced by V' .

The stored energy $E(V')$ is found by integration of $p dV$ (for reasons of simplicity only the interesting case where $V' > Al$ is considered):

$$\begin{aligned}
 E(V') &= \int_0^{Al} \frac{\rho g}{A} V dV + \int_{Al}^{V'} \rho g \left(l + \frac{V - Al}{A - a} \right) dV = \\
 &= \frac{\rho g}{A} \left(\frac{V^2}{2} \Big|_0^{Al} \right) + \rho g \left(V l \Big|_{Al}^{V'} + \frac{V^2}{2(A-a)} \Big|_{Al}^{V'} - \frac{V Al}{A-a} \Big|_{Al}^{V'} \right) \\
 &= \frac{\rho g}{A} \frac{(Al)^2 - 0}{2} + \rho g \left(V l - Al^2 + \frac{V'^2 - (Al)^2}{2(A-a)} - \frac{(V' - Al)Al}{A-a} \right) = \\
 &= \frac{\rho g}{2(A-a)} (V'^2 - 2V'al + aAl^2) \quad \text{for } V' > Al
 \end{aligned}$$

Note the importance of the separating the two integration intervals for finding the correct expression for the energy.

$$E(V') = \frac{\rho g V'^2}{2A} \quad \text{for } V' < Al$$

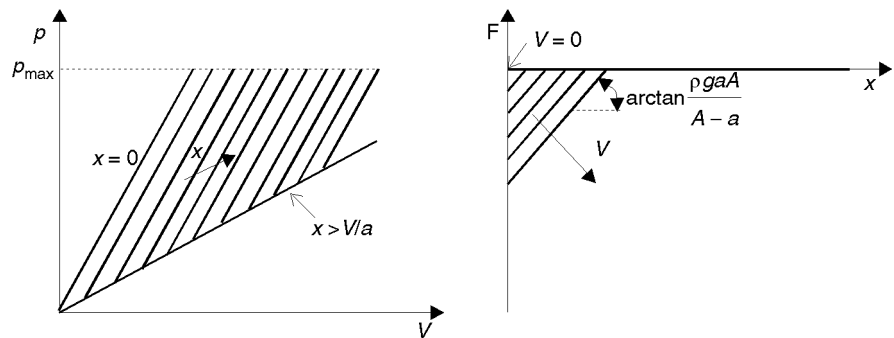
- 18.8 In the p - V characteristic the starting point of the second, steeper part of the characteristic at a specific value of V depends on the value of x . For p an upper bound always exist in the form of the rim of the water container, for example at level h [m]. In the F - x characteristic the force should be zero as long as $V < Ax$ and x should be larger than zero. F is negative, but its derivative with respect to x is positive, viz.

$$\frac{\partial F}{\partial x} = \frac{\rho g a A}{A - a} \quad \text{for } V > Ax \text{ or } x < V/A$$

so

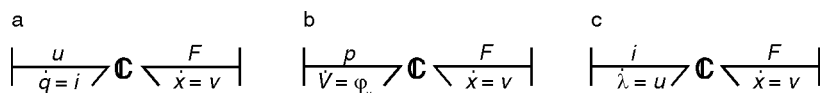
$$\frac{\partial F}{\partial x} = 0 \quad \text{for } x > V/A$$

Note the parallel characteristics with slope $\rho g a A / (A - a)$ in the fourth quadrant that coincide with the x -axis after reaching it.



18.9 This is the case when the plates are connected elastically to each other, for example via an elastic dielectric (material between the plates with a specific dielectric constant) or an elastic ‘suspension’ (membrane). Note that this is, and should be, the case for all realizations (electret microphone, electrostatic loudspeaker). If there would not be an elastic suspension of the moving plate, it will irrevocably be drawn onto the other plate, such that the two plates cease being a capacitor.

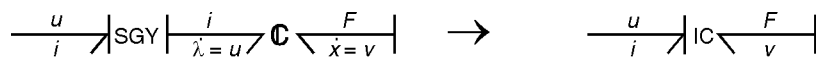
18.10 The required bond graphs are shown in the next figure:



Three examples of two-port C's:

- a capacitor with moving plate
- b water container with float
- c elastic coil

In case c the magnetic port becomes a dual storage element (I-type) in the electric domain. In other words: the electric and magnetic domains are connected by a unit gyrator (SGY). This two-port can thus be represented as follows too:



Two-port C represented by an I-type port in the electric domain, coupled to a C-type port in the mechanical domain

Note that a similar representation of the partial dualization of the ports of an n -port storage element with $n > 2$ is more cumbersome.

18.11 This problem can be solved in different ways. In this case it is obvious to sum the kinetic energy of the system with the length of the spring $x = l$ fixed, viz.

$$E(b) = \frac{b^2}{2J} = \frac{b^2}{2ml^2}$$

and the elastic energy of the spring

$$E(x) = \frac{Kl^2}{2}$$

and next to replace the parameter l in the first expression by the state variable x as shown before in the examples treated earlier.

$$E(b, x) = \frac{b^2}{2mx^2} + \frac{Kx^2}{2}$$

Partial differentiation of this expression results in the constitutive relations of the two-port storage element with ports $(\omega, db/dt)$ and $(F, dx/dt)$, so

$$\omega(b, x) = \frac{b}{mx^2}$$

$$F(b, x) = \frac{-b^2}{mx^3} + Kx$$

Next repeated partial differentiation results in the constitutive matrix in an operating point (x_0, b_0)

$$\begin{pmatrix} \frac{1}{mx_0^2} & \frac{-2b_0}{mx_0^3} \\ \frac{-2b_0}{mx_0^3} & \frac{+3b_0^2}{mx_0^4} + K \end{pmatrix}$$

This is a symmetric matrix again. Its determinant equals

$$\frac{3b_0^2}{m^2x_0^6} + \frac{K}{mx_0^2} - \frac{4b_0^2}{m^2x_0^6} = \frac{1}{mx_0^2} \left(K - \frac{b_0^2}{mx_0^4} \right)$$

Note that $K > b_0^2/mx_0^4$ is the condition for intrinsic stability.

In order to get a better insight in its meaning this condition is rewritten in terms of ω_0 :

$$\frac{K}{m} > \omega_0^2 \quad \text{where} \quad \omega_0 = \frac{b_0}{mx_0^2}$$

In other words: at a specific stiffness of the spring and a specific mass the angular velocity has an upper bound. This means that if the mass is rotated faster and faster, the spring would become infinitely long. In case of ‘Hookian’ spring this is possible, in practice it is not, due to the fact that all real springs saturate, i.e. reach an extension where the stiffness increases (such that the intrinsic stability condition is again satisfied) or where the spring breaks (a ‘catastrophic’ instability).

- 18.12 Substitution of $\lambda = \frac{A\mu}{x}i$ into $F(\lambda, x)$ according to (18.22) results in:

$$F(i, x) = \frac{A\mu}{x^2}i^2 + K(x-l)$$

Substitution of $\lambda = \frac{A\mu}{x}i$ in $E(\lambda, x)$ according to (18.20) results in:

$$E(i, x) = \frac{A\mu}{2x}i^2 + \frac{K(x-l)^2}{2}$$

For $F(i, x) = \frac{\partial E(i, x)}{\partial x}$ it holds that: $F(i, x) = -\frac{A\mu}{x^2}i^2 + K(x-l)$

This expression for $F(i, x)$ differs from the expression found earlier by the minus sign of the first term.

- 18.13 Only consider the case that $V > Ax$, because there is no two-port storage element otherwise.

$$E(V, x) = \frac{\rho g}{2(A-a)}(V^2 - 2Vax + aAx^2) \quad \text{for } V > Ax$$

$$F(V, x) = \frac{\partial E(V, x)}{\partial x} = \frac{-\rho ga}{A-a}(V - Ax) \quad \text{for } V > Ax$$

substitution of V by

$$V(p, x) = \frac{(A-a)p}{\rho g} + ax$$

results in:

$$E(p, x) = \frac{(A-a)p^2}{2\rho g} + \frac{\rho g ax^2}{2}$$

and

$$F(p, x) = -a(p - \rho gx)$$

If the partial derivative of E with respect to x is determined

$$\frac{\partial E(p, x)}{\partial x} = \rho g ax$$

it can be observed that in this case it does *not* hold that

$$F(p, x) = \frac{-\partial E(p, x)}{\partial x}$$

but

$$F(p, x) = -pa + \frac{\partial E(p, x)}{\partial x}$$

18.14 Inspection of the relations (18.6), (18.7), (18.13) mentioned earlier and (18.14), (18.21) and (18.22) shows this immediately.

18.15 This form does not correspond to the preferred causality as λ is a variable of the generalized q -type and i a generalized effort (of the magnetic domain), so

$$\mathbf{q} = \begin{pmatrix} L_1 & M \\ M & L_2 \end{pmatrix} \mathbf{e}$$

Its preferred causality is

$$\mathbf{e} = \begin{pmatrix} L_1 & M \\ M & L_2 \end{pmatrix}^{-1} \mathbf{q} = \begin{pmatrix} L_1 L_2 - M^2 \end{pmatrix}^{-1} \begin{pmatrix} L_2 & -M \\ -M & L_1 \end{pmatrix} \mathbf{q}$$

18.16 Chapter 17 showed that for an ideal gas it holds that:

$$T(v, s) = T_0 \left(\frac{v}{v_0} \right)^{\frac{-R}{c_v}} e^{\frac{s-s_0}{c_v}} \quad (17.66)$$

where $v = V/N$ and $s = S/N$ and

$$p(v, s) = p_0 \left(\frac{v}{v_0} \right)^{\frac{-R}{c_v}-1} e^{\frac{s-s_0}{c_v}} \quad (17.67)$$

where s and v are the values of S and V for $N = 1$ mole, such that

$$\frac{\partial T}{\partial V} = \frac{1}{N} \frac{\partial T}{\partial v} = \frac{-RT_0}{Nv_0 c_v} \left(\frac{v}{v_0} \right)^{\frac{-R}{c_v}-1} e^{\frac{s-s_0}{c_v}} = \frac{-p}{Nc_v}$$

and

$$\frac{\partial p}{\partial S} = \frac{1}{N} \frac{\partial p}{\partial s} = p_0 \left(\frac{v}{v_0} \right)^{\frac{-R}{c_v}-1} \frac{e^{\frac{s-s_0}{c_v}}}{Nc_v} = \frac{p}{Nc_v}$$

Recall that

$$\frac{\partial E(V, S)}{\partial V} = -p$$

and

$$\frac{\partial E(V, S)}{\partial S} = T$$

such that Maxwell reciprocity means that

$$\frac{\partial T}{\partial V} = \frac{\partial(-p)}{\partial S} = \frac{-\partial p}{\partial S} = \frac{-p}{Nc_v}$$

- 18.17 The coupling factor of the flat-plate capacitor with moving plate is infinite when there is no elastic suspension (the capacitor is not intrinsically stable in that case). If there would be a suspension with spring constant K the coupling factor is:

$$k = q_0 \frac{1}{\sqrt{Kx_0 A \epsilon}}$$

The coupling factor of the water container with float is:

$$k = \sqrt{\frac{a}{A}}$$

and of the elastic coil

$$k = \lambda_0 \sqrt{\frac{A\mu}{Kx_0}}$$

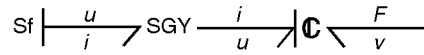
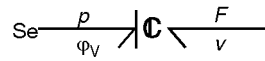
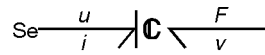
- 18.18 For k it is found that:

$$k = \left[\frac{M^2}{L_1 L_2} \right]^{0.5}$$

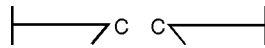
such that intrinsic stability means that:

$$|M| < (L_1 L_2)^{0.5} \quad L_1 > 0 \text{ and } L_2 > 0$$

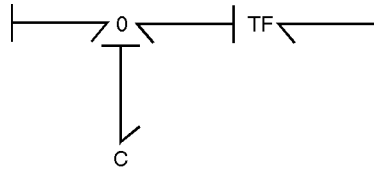
- 18.19 In all cases it concerns a change of causality, viz. of integral into differential causality of the non-mechanical port, so the electric, hydraulic and magnetic ports respectively:



For $k = 0$ the two-port C-type storage elements change in all cases into two one-port C-type storage elements:



For $k = 1$ a junction structure emerges with one one-port C-type storage element connected to a 0-junction, that is directly connected to one of the ports and to the other port via a transformer:



18.20 Application of Euler's theorem to the function

$$F(\mathbf{x}) = F(x, y, z) = 2xyz^2 + 3x^2y^2 + x^3z \quad \text{where } \mathbf{x} = x, y, z \quad (n = 4, k = 3)$$

results in:

$$\frac{\partial F(x)}{\partial x} x = 2xyz^2 + 6x^2y^2 + 3x^3z$$

$$\frac{\partial F(x)}{\partial y} y = 2xyz^2 + 6x^2y^2 + 0$$

$$\frac{\partial F(x)}{\partial z} z = 4xyz^2 + 0 + x^3z$$

$$\sum_{i=1}^k \frac{\partial F(x)}{\partial x_i} x_i = 8xyz^2 + 12x^2y^2 + 4x^3z = 4F(x)$$

18.21 For the linear constitutive relations it holds that $\mathbf{e}(\mathbf{q}) = \mathbf{A}\mathbf{q}$ with $\mathbf{A}$ a constant matrix, such that $\mathbf{e}(\mathbf{a}\mathbf{q}) = \mathbf{a}\mathbf{A}\mathbf{q}$. In other words: the constitutive relations are homogenous functions of the first degree. Based on (18.63) this means that the energy function is a homogenous function of the second degree. This corresponds to the quadratic form of the energy function in that case.

18.22 For the generalized energy function it holds that:

$$dE(q) = edq$$

In case a p -type storage element is considered, like a mass, this can be written:

$$dE(p) = fdp = vdp \quad \text{where } v = \frac{1}{m} p$$

It follows for the kinetic energy:

$$E = \frac{1}{2m} p^2$$

If e and q —or $f (= v)$ and p — are exchanged, a Legendre transform is performed in principle and the co-energy is obtained:

$$dE^*(v) = p df = p dv \quad \text{where } p = mv$$

This results in an expression for the kinetic co-energy:

$$E^* = \frac{1}{2} mv^2$$

- 18.23 The change of causality of the electric port of the capacitor, causing the voltage u to be imposed, means that a Legendre transform is to be performed (with minus sign according to (18.102)) with respect to the charge q in order to find the co-energy.

For the Legendre transform of $E(q, x)$ with respect to q it holds that:

$$\mathcal{L}\{E(q, x)\} \Big|_{q(u, x)} = E(q, x) - u(q, x)q.$$

Substitution of $E(q, x) = \frac{q^2 x}{2A\epsilon}$ and $u(q, x) = \frac{qx}{A\epsilon}$ results in:

$$\mathcal{L}\{E(q, x)\} \Big|_{q(u, x)} = -\frac{q^2 x}{2A\epsilon} (= -E(q, x)).$$

and substitution of $q(u, x) = \frac{A\epsilon u}{x}$:

$$\mathcal{L}\{E(q, x)\} \Big|_{q(u, x)} = -\frac{A\epsilon u^2}{2x} (= -E^*(u, x)).$$

For the force $F(u, x)$ holds that:

$$F(u, x) = \frac{\partial}{\partial x} \mathcal{L}\{E(q, x)\} \Big|_{q(u, x)} = \frac{A\epsilon u^2}{2x^2} \quad (= \frac{\partial}{\partial x} \{-E^*(u, x)\})$$

Comparison of this result with (18.26) shows that the unjustified minus sign that was found there has disappeared accordingly. Consequently, back substitution of

$$u = \frac{qx}{A\epsilon}$$

results in the correct expression for F found earlier (18.7). In section 2 (after (18.26)) it was concluded that $E(q,x)$ and $E(u,x)$ would not be identical. By contrast, it can be concluded now that $E(q,x)$ and $-E^*(u,x)$ are indeed identical.

The change of causality of the electric port of the coil to which the current i is imposed, means that a Legendre transform (with minus sign) with respect to the flux linkage λ should be performed in order to find the co-energy. It holds for the Legendre transform of $E(\lambda,x)$ with respect to λ :

$$\mathcal{L}\{E(\lambda,x)\}\big|_{\lambda(i,x)} = E(\lambda,x) - i(\lambda,x)\lambda$$

Substitution of $E(\lambda,x) = \frac{\lambda^2 x}{2A\mu} + \frac{K(x-l)^2}{2}$ and $i(\lambda,x) = \frac{x\lambda}{A\mu}$ results in:

$$\mathcal{L}\{E(\lambda,x)\}\big|_{\lambda(i,x)} = -\frac{\lambda^2 x}{2A\mu} + \frac{K(x-l)^2}{2}$$

and substitution of $\lambda(i,x) = \frac{A\mu i}{x}$:

$$\mathcal{L}\{E(\lambda,x)\}\big|_{\lambda(i,x)} = -\frac{A\mu i^2}{2x} + \frac{K(x-l)^2}{2} \quad (= -E^*(i,x))$$

For the force $F(i,x)$ holds that:

$$F(i,x) = \frac{\partial}{\partial x} \mathcal{L}\{E(\lambda,x)\}\big|_{\lambda(i,x)} = \frac{A\mu i^2}{2x^2} + K(x-l) \quad (= \frac{\partial}{\partial x} \{-E^*(i,x)\})$$

Here the (unjustified) minus sign found in assignment 18.12 has disappeared, corresponding to equation (18.22). Conclusion: $E(\lambda,x)$ is identical to $-E^*(i,x)$.

18.24 Only consider the situation where $V > Ax$:

$$E(V,x) = \frac{\rho g}{2(A-a)} \left(V^2 - 2Vax + aAx^2 \right)$$

For the Legendre transform of $E(V,x)$ with respect to V it holds that:

$$\mathcal{L}\{E(V,x)\}\big|_{V(p,x)} = E(V,x) - p(V,x)V$$

Substitution of

$$E(V,x) = \frac{\rho g}{2(A-a)} \left(V^2 - 2Vax + aAx^2 \right) \text{ for } V > Ax \text{ according to (18.11)}$$

and

$$p(V, x) = \frac{\rho g}{A-a}(V - ax) \text{ for } V > Ax \text{ according to (18.14a),}$$

results in:

$$\begin{aligned} \mathcal{L}\{E(V, x)\}|_{V(p, x)} &= \frac{\rho g}{A-a} \left\{ \frac{1}{2} V^2 - Vax + \frac{1}{2} aAx^2 - V^2 + Vax \right\} = \\ &= \frac{\rho g}{A-a} \left\{ -\frac{1}{2} V^2 + \frac{1}{2} aAx^2 \right\} \end{aligned}$$

Substitution of $V(p, x) = \frac{(A-a)p}{\rho g} + ax$ next results in:

$$\begin{aligned} \mathcal{L}\{E(p, x)\}|_{V(p, x)} &= \frac{\rho g}{A-a} \left\{ -\frac{1}{2} \left(\frac{(A-a)p}{\rho g} + ax \right)^2 + \frac{1}{2} aAx^2 \right\} = \\ &= -\frac{(A-a)p^2}{2\rho g} - apx + \frac{1}{2} \rho g ax^2 \quad (= -E^*(p, x)) \end{aligned}$$

$$\text{so } E^*(p, x) = \frac{(A-a)}{2\rho g} p^2 + apx - \frac{1}{2} \rho g ax^2$$

such that $F(p, x)$ can be found from $E^*(p, x)$ as follows:

$$F(p, x) = -\frac{\partial E^*(p, x)}{\partial x} = -ap + \rho g ax = -a(p - \rho g x)$$

thus resulting on the correct answer this time.

2 Solutions of the self test

- 1 If a storing port does not have integral causality, the co-energy in stead of the energy has to be derived. The co-energy is found by a Legendre transform: the co-energy $E^*(e)$ equals the negative value of the Legendre transform of $E(q)$ with respect to q . The Legendre transform is given by the definition in equation (18.84).
- 2 The energy stored in the flat-plate capacitor in figure 18.1 is given by (18.3):

$$E(q, x) = \frac{q^2}{2C(x)} = \frac{q^2 x}{2A\epsilon}$$

If q and x are multiplied by a factor a , it is found that

$$E(aq, ax) = \frac{a^2 q^2}{2C(ax)} = \frac{a^3 q^2 x}{2A\epsilon} = a^3 \frac{q^2 x}{2A\epsilon} = a^3 E(q, x)$$

Therefore, this function is a homogenous function of the third degree.

- 3 The homogeneity of the functions is checked by multiplication of the variables with a factor α :

$$E(\alpha q) = a e^{b\alpha q} \neq \alpha^? E(q)$$

$$E(\alpha q_1, \alpha q_2) = a e^{b \frac{\alpha q_1}{\alpha q_2}} = a e^{b \frac{q_1}{q_2}} = \alpha^0 E(q_1, q_2)$$

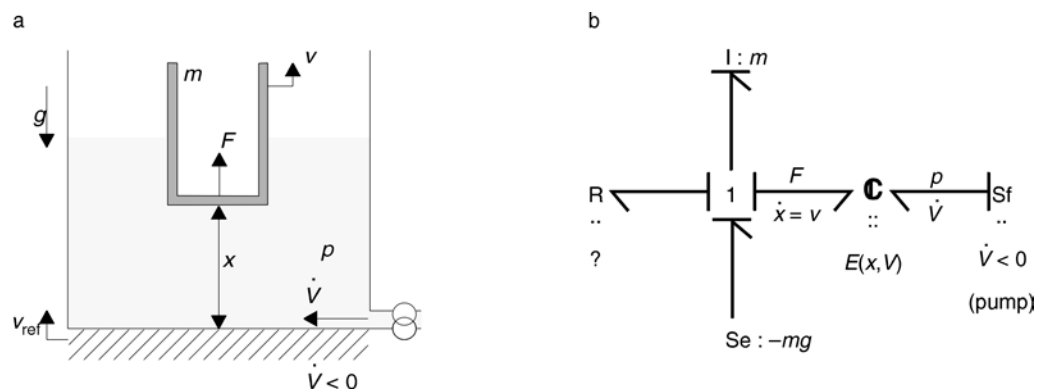
From this follows that the first function is inhomogeneous and the second homogenous of the zeroth degree.

- 4 If all capacitor equations and coil equations are linear, such that the energy $E(\mathbf{q}, \boldsymbol{\lambda})$ is equal in value to the co-energy $E^*(\mathbf{u}, \mathbf{i})$.
- 5 Apart from the examples of a lever, a capacitor with moving plate like in a electrostatic microphone and a water container with float like in a cistern discussed in the text, the following examples are mentioned too: a piezo-electric gas lighter, a cylinder-piston combination filled with compressible oil, a bending beam clamped at one side, a coil with moving iron core (LVDT = linear variable displacement transmitter), a voice coil-magnet-combination in an electrodynamic loudspeaker, et cetera.
- 6 The first situation is shown in the following figures. In this case the two-port C has integral causality, such that the relations can be derived from the energy function. The result equals what is found earlier in section 2:

$$F(V, x) = \frac{\partial E}{\partial x} = \frac{-Va\rho g}{(A-a)} + \frac{aAx\rho g}{(A-a)} = -\frac{\rho ga}{A-a}(V - Ax) \quad \text{for } V > Ax$$

$$p(V, x) = \frac{\partial E}{\partial V} = \frac{V\rho g}{(A-a)} - \frac{ax\rho g}{(A-a)} = \frac{\rho g}{A-a}(V - ax) \quad \text{for } V > Ax$$

The situation $V < Ax$ will not often occur in case of a ship in a dock.



Ship in a dock that is emptied by a pump

a IPM

b Bond graph

The second situation is shown in the following figures. In this case the causality of the p - dV/dt port changes in differential causality, thus requiring Legendre transformation of the energy with respect to V in order to derive the relations. The result corresponds to the result in the solution of assignment 18.24

$$E^*(p, x) = \frac{-\rho g a x^2}{2} + \frac{(A-a)p^2}{2\rho g} + p a x$$

gives

$$F(p, x) = \frac{-\partial E^*}{\partial x} = \rho g a x - p a = -(p a - \rho g a x)$$

and

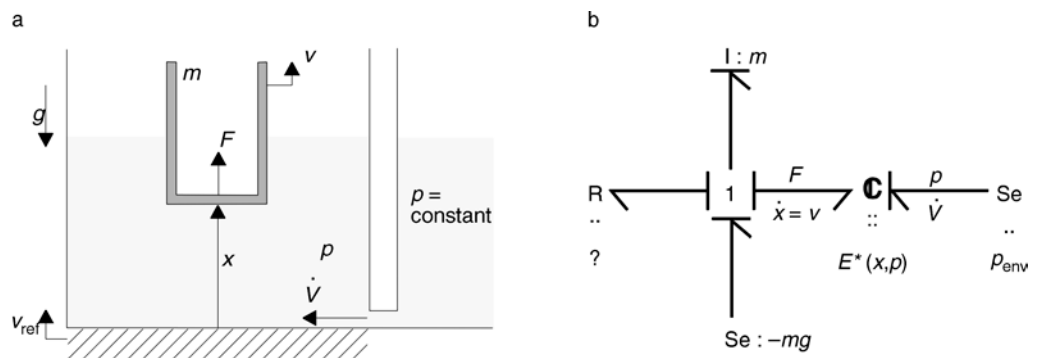
$$V(p, x) = \frac{\partial E^*}{\partial p} = + \frac{(A-a)p}{\rho g} + a x$$

In general it holds that:

$$V = -\frac{\partial E}{\partial p}.$$

This means that V and p are inversely proportional. Hence:

$$V(p, x) = -\frac{\partial}{\partial p}(-E^*) = \frac{\partial E^*}{\partial p}$$



Ship in a dock in open connection to the harbor

a IPM

b Bond graph